

# RAG evaluation checklist

The checks worth running before a retrieval system goes in front of users. Not exhaustive - ordered by how often skipping one has caused a visible failure.

## 1. Before you measure anything

- ☐ Do you have a test set of real questions, written by someone who is not you?
- ☐ Does it include questions the corpus genuinely cannot answer? If not, you cannot measure refusal.
- ☐ Is there a known-correct passage for each question, so retrieval can be scored separately from generation?

## 2. Retrieval

- ☐ Recall@k on your test set, at the k you actually pass to the model.
- ☐ What fraction of queries retrieve zero relevant passages? That number is your ceiling.
- ☐ Does hybrid search beat dense alone on your data? Check rather than assume.
- ☐ Are chunk boundaries splitting tables, code blocks or numbered lists mid-structure?
- ☐ Does reranking change the top-k enough to justify the latency?

## 3. Generation

- ☐ Is every factual claim traceable to a retrieved passage?
- ☐ When retrieval returns nothing useful, does the system say so - or invent an answer?
- ☐ Does answer quality collapse when the relevant passage sits in the middle of a long context?
- ☐ Are citations pointing at the passage that actually supports the sentence, or just at something nearby?

## 4. The seams

- ☐ What happens on a question spanning two documents that contradict each other?
- ☐ What happens when the corpus is updated but the index is not?
- ☐ Is there a permission boundary in the corpus? If so, is it enforced at retrieval - not in the prompt?
- ☐ What does a malicious question retrieve? Try to get it to surface something it should not.

## 5. Operations

- ☐ Cost per query at expected volume, including reranking and any repeated calls.
- ☐ p95 latency, measured end to end rather than per component.
- ☐ Are queries and retrieved passages logged so a bad answer can be reconstructed later?
- ☐ Is there a regression suite that runs before each index or prompt change?

## **The one that catches most people**

Eyeballing twenty answers and concluding it works. Twenty is enough to see that it is not broken; it is nowhere near enough to see the shape of what it gets wrong. Build the test set first - it is a day of work and it is the difference between improving the system and guessing at it.

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